

Project#5

CUDA: Monte Carlo Simulation

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1. Tell what machine you ran this on

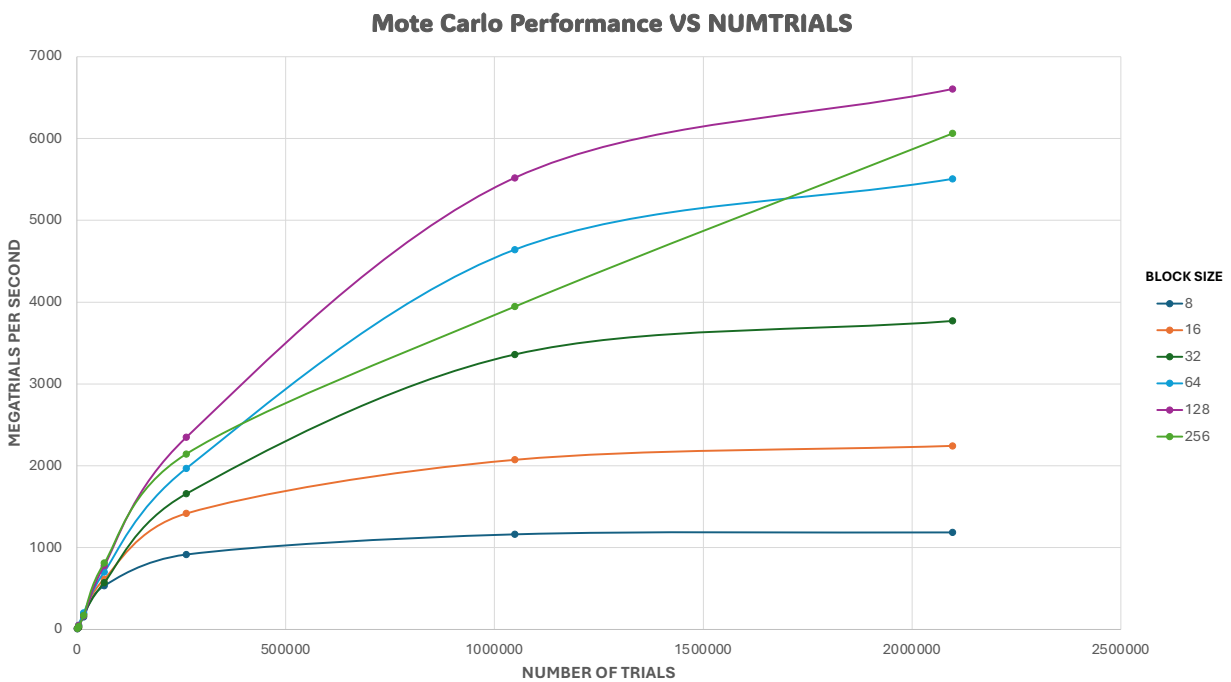
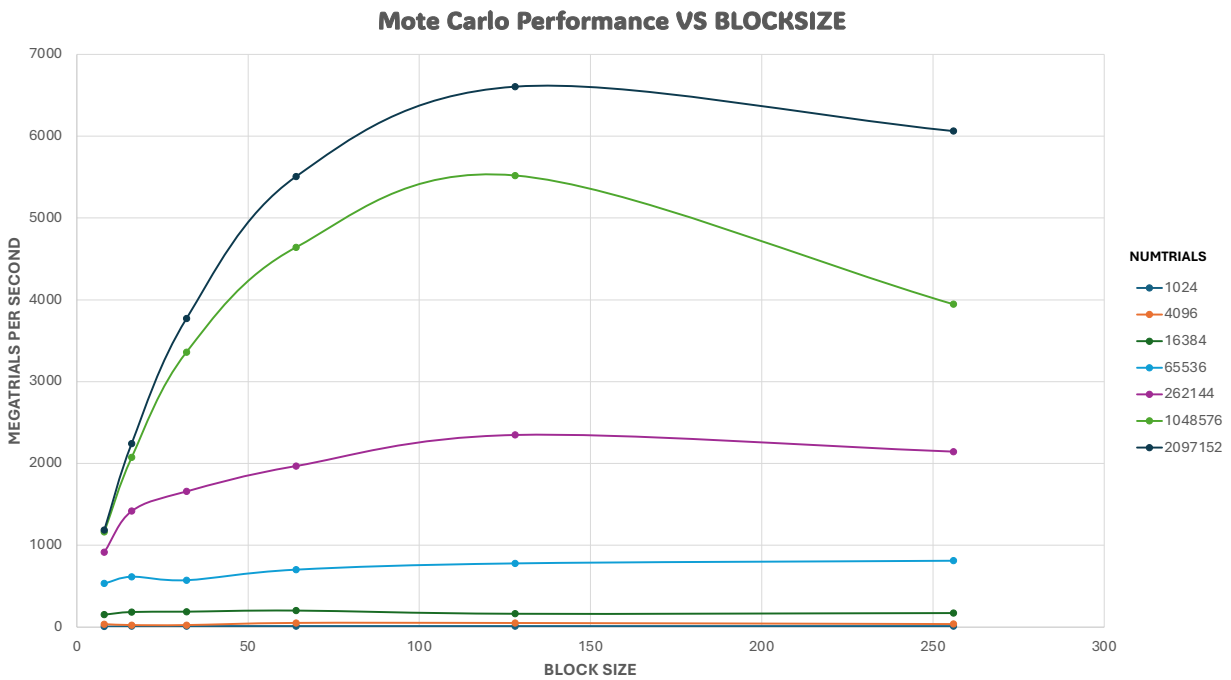
- I ran this on rabbit server.

2. What do you think this new probability is?

- The probability is about 83.8%.

3. Show the rectangular table and the two graphs.

NUMTRIALS	BLOCKSIZE	MegaTrialsPerSec	Probability
1024	8	6.7797	83.89
1024	16	10.306	82.32
1024	32	12.6432	84.86
1024	64	12.9502	83.98
1024	128	12.7592	83.11
1024	256	12.7592	83.3
4096	8	36.3843	83.2
4096	16	25.3817	83.08
4096	32	25.3415	83.35
4096	64	51.6963	84.62
4096	128	50.6931	83.89
4096	256	37.636	84.3
16384	8	152.6536	83.94
16384	16	184.9711	83.92
16384	32	188.9299	84.28
16384	64	202.4516	84.28
16384	128	163.2653	83.66
16384	256	171.9852	84.06
65536	8	533.3333	83.94
65536	16	615.7547	83.9
65536	32	573.6695	84.14
65536	64	702.5729	83.77
65536	128	779.0034	84.01
65536	256	811.4105	83.96
262144	8	915.9213	83.9
262144	16	1417.3011	83.88
262144	32	1657.9639	83.86
262144	64	1968.7576	83.71
262144	128	2348.6238	83.86
262144	256	2145.0642	83.76
1048576	8	1162.5629	83.8
1048576	16	2073.2679	83.8
1048576	32	3360.8204	83.82
1048576	64	4642.0173	83.86
1048576	128	5520.2156	83.85
1048576	256	3945.5749	83.83
2097152	8	1186.1934	83.82
2097152	16	2243.8456	83.79
2097152	32	3770.7712	83.8
2097152	64	5507.2268	83.81
2097152	128	6605.7855	83.81
2097152	256	6062.5348	83.8



4. What patterns are you seeing in the performance curves?

- In “Performance vs BlockSize” graph as the block size increases, there is a noticeable improvement in performance. However, after the block size exceeds 128, the performance shows slight decline.

- In “Performance vs NumTrials” graph, as the trials increase, performance increases as well. The graph with Block size 256 shows most linear like graph.

5. Why do you think the patterns look this way?

- The patterns observed in the performance curves in “Performance vs BlockSize” graph are likely due to the increased efficiency gained from parallel processing as the number of trials and block sizes increase which increases threads number.
- Larger block sizes can better utilize the GPU's parallel processing capabilities up to an optimal point. Beyond this point, performance gains diminish due to overheads or limitations in the hardware architecture.

6. Why is a BLOCKSIZE of 8 so much worse than the others?

- This is because with only 8 blocks there is not much thread to work. Similar for 16 Block Size
- With less than 32 blocks, the GPU's threads are underutilized.
- As learned in class, setting 32 blocks is not an ideal choice when there are 32 GPU cores. The best performance is achieved with a block size of $2 \times \text{CoreNum}$ or $3 \times \text{CoreNum}$
- This is because extra threads are needed to swap in when existing threads are waiting to fetch data from memory.

7. How do these performance results compare with what you got in Project #1? Why?

- The maximum performance results are approximately 2000 times faster compared to the 8-thread calculation from Project #1 and 128 block size from Project #5.
- In project #1, it used CPU to make calculation parallel maximum of 8 threads (Since there are only 8-cpu-core in server), but in Project #5 we have utilized more threads by using up to 256 blocks.

8. What does this mean for what you can do with GPU parallel computing?

- The results indicate that GPU parallel computing is highly efficient for tasks that can be decomposed into multiple parallel threads. This approach shows the full potential of multi-threaded computing, resulting in significant performance gains for suitable applications. Tasks involving large-scale computations can be executed much more efficiently on a GPU compared to traditional CPU-based processing.